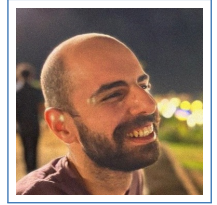


Mahran Shehadeh

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Single-molecule biophysicist — photon-level optical instrumentation, nanophotonic enhancement, and computational photon-stream analysis. Interdisciplinary background across physics, materials science, and biophysics.

Education

- 2021–2025 **Ph.D. in Biophysics**, *Weizmann Institute of Science*, Rehovot, Israel
Department of Chemical and Biological Physics.
Advisor: Prof. Gilad Haran.
- 2016–2019 **M.Sc. in Physics**, *Weizmann Institute of Science*, Rehovot, Israel
Department of Particle Physics.
Advisors: Prof. Eilam Gross and Prof. Ehud Duchovni.
- 2012–2016 **B.Sc. in Physics and Materials Science & Engineering**, *Technion — Israel Institute of Technology*, Haifa, Israel

Research Experience

- 2021–2025 **Ph.D. Researcher — Single-Molecule Biophysics**, *Weizmann Institute of Science*, Rehovot, Israel
Advisor: Prof. Gilad Haran. Plasmon-enhanced single-molecule FRET to resolve structural transitions and dynamics in proteins, across several connected sub-projects:
- Nanofabrication of plasmonic structures — nanoholes and bowtie nanoantennas — by focused-ion-beam (FIB) and electron-beam lithography (EBL), with SEM characterization.
 - Surface passivation and sample chemistry to enable clean single-molecule measurements on these structures.
 - Computational electromagnetic simulation (FDTD and BEM) to optimize geometries for near-field enhancement and emission control.
 - Study of fluorophore behavior in confined, enhanced fields (photophysics, FRET, photon yield), analyzed with custom photon-stream pipelines using Hidden Markov Models, deep learning, and Bayesian inference.
- 2019–2020 **Research Assistant — Quantum Optics / Light–Matter Interactions in Ultracold Atoms**, *Weizmann Institute of Science*, Rehovot, Israel
Group of Prof. Nir Davidson.
Investigated nonlinear optical phenomena and collective light–matter coupling in Bose–Einstein condensates.
- 2016–2019 **M.Sc. Research — High-Energy Physics (ATLAS Experiment)**, *Weizmann Institute of Science*, Rehovot, Israel
Developed ROOT/C++ software for ATLAS data processing and statistical analysis; contributed to collaborative efforts in searches for new physics and precision Standard Model measurements.
- 2016 **Summer Research Project — Astroparticle and Dark Matter Physics**, *Weizmann Institute of Science*, Rehovot, Israel
Contributed to the development of a multi-ton liquid xenon time-projection chamber for rare-event detection, focusing on low-light and low-charge signal characterization.

- 2015 **Computational Physics and Chemistry Project**, *Technion — Israel Institute of Technology*, Haifa, Israel
Advisor: Prof. Maytal Caspary Toroker.
Modeled transition-metal oxide catalysts for photoelectrochemical water splitting using DFT (VASP, Quantum ESPRESSO).

Publications

- 2017 Zaffran, J., Stevens, M. B., Trang, C. D. M., Nagli, M., **Shehadeh, M.**, Boettcher, S. W., & Caspary Toroker, M. (2017). Influence of Electrolyte Cations on Ni(Fe)OOH-Catalyzed Oxygen Evolution Reaction. *Chemistry of Materials*, 29(11), 4761–4767.
- 2018 Zaffran, J., Nagli, M., **Shehadeh, M.**, & Caspary Toroker, M. (2018). Efficient Cationic Agents for Exfoliating Two-Dimensional Nickel Oxide Sheets. *Theoretical Chemistry Accounts*, 137(1), 1–5.
- 2021 Jakub Jungwirth, Sayan Mondal, **Mahran Shehadeh**, & Gilad Haran. (2021). Single-Molecule Insights into Allosteric Regulation of the Nuclear Receptor Protein RXR. *Biophysical Journal*.
- 2025 **Manuscript in preparation**, *Toward Understanding Structural Transitions and Dynamics in Proteins Using Plasmon-Enhanced Single-Molecule FRET*.

Consortium Publications (ATLAS Collaboration)

- 2019-2023 **CERN–LHC–ATLAS**; see INSPIRE-HEP profile: inspirehep.net/authors/1624629.

Conferences, Internships, and Schools

- 2025 **Biophysical Society Annual Meeting**, Los Angeles: *Toward Understanding Structural Transitions and Dynamics in Proteins Using Plasmon-Enhanced Single-Molecule FRET*.
- 2024 **Weizmann AI Hub Internship** — Developed LSTM models for photon time-series analysis in single-molecule FRET experiments.
- 2021 **Summer School on Surface Plasmons**, Institute for the Science of Light, Paris-Saclay.

Scholarships and Awards

- 2026 The Bina Innovation Fellowship (3rd Cohort), An industry-academy program for outstanding PhD students.
- 2025 Breakthrough Prize in Fundamental Physics (ATLAS Collaboration).
- 2023 Certificate of Appreciation for Promoting Diversity and Inclusion, Weizmann Institute of Science.
- 2016 Council for Higher Education Scholarship for Outstanding Students from the Arab Community.
- 2015 Amos de-Shalit Ulpana Scholarship for Physics, Weizmann Institute of Science.
- 2013 Technion Dean's Prize for Academic Excellence.

Teaching and Outreach

- 2022–2023 Instructor of mathematics, critical thinking, and English in Arab secondary schools (Lod and Ramle).
- 2021 Founder, Arab Forum at the Weizmann Institute — Promoting well-being, culture, and inclusion.
- 2018–Present Science tutor, Davidson Institute of Science Education — Outreach for high-school and pre-academic students.

- 2018–2022 Volunteer teacher of physics and mathematics, Lod High School (Periphery Program).
2015–2016 Teaching assistant, First-Year Physics Laboratory, Technion.
2015–2017 Tutor, Virtual School (") — Teaching advanced physics (5 units).
2013–2014 Participant, Perach Mentoring Program — Personal mentoring and educational support.
2012–2015 Translator, Khan Academy — English, Hebrew, and Arabic content. [khanacademy.org](https://www.khanacademy.org)
2011–2012 Personal tutor, Ma'an Foundation, Makr.

Certifications

Technion Courses Teaching Certificate.
ITTT 120-hour TESOL/TEFL Certificate; 50-hour Teaching Business English Certificate.
365 Data Science — Certificate of Achievement: Introduction to Data and Data Science.
Professional Acting Certificate (1-year program), Jaffa Arab-Hebrew Theatre.

Skills

- | | |
|-------------------------|--|
| Optical instrumentation | Confocal fluorescence microscopy, single-photon detection (photon counting / time-resolved photon streams), single-molecule FRET, plasmonic / nanophotonic field enhancement, optical setup design and alignment. |
| Fabrication & materials | FIB, EBL, SEM, atomic-layer deposition (ALD); plasmonic nanostructure fabrication and surface passivation; materials-science foundation (B.Sc.). |
| Computation & modeling | Photon time-series analysis (Hidden Markov Models, LSTM/deep learning, Bayesian inference); electromagnetic simulation (FDTD, BEM); DFT (VASP, Quantum ESPRESSO); Python, MATLAB, Mathematica, C/C++, C#, ROOT, Linux. |
| Languages | Arabic (Native), English (Fluent), Hebrew (Fluent); Spanish (beginner). |

Academic References

Due to confidentiality policies, formal recommendation letters can be provided immediately upon request or sent directly to the hiring committee.

1. Prof. Gilad Haran (Ph.D. Advisor)

Department of Chemical and Biological Physics
Weizmann Institute of Science, Rehovot, Israel
Email: gilad.haran@weizmann.ac.il
Relationship: Ph.D. Thesis Advisor

2. Prof. Hagen Hofmann

Department of Chemical and Structural Biology
Weizmann Institute of Science, Rehovot, Israel
Email: hagen.hofmann@weizmann.ac.il
Relationship: Ph.D. Committee Member & Expert in Single-Molecule Protein Dynamics

3. Prof. Adi Salomon

Department of Chemistry & Institute of Nanotechnology (BINA)
Bar-Ilan University, Ramat Gan, Israel
Email: adi.salomon@biu.ac.il
Relationship: Ph.D. Committee Member & Expert in Nanophotonics/Plasmonics